

12. Implement Iris Flower Classification using KNN

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42)

k = 3

model = KNeighborsClassifier(n_neighbors=k)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Labels:", y_test)

print("Predicted Labels:", y_pred)

print("Accuracy:", accuracy_score(y_test, y_pred))

sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("Predicted Flower:", iris.target_names[prediction][0])
```

OUTPUT:

```
Actual Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Predicted Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
Accuracy: 1.0
Predicted Flower: setosa
```